

Integrative analysis for sc-/sn-RNA-Seq Analysis in 10X Genomics data

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```
## The following objects are masked _by_ .GlobalEnv:
##
##     integration_method, root_dir
```

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DNB Bioinformatics Core Pipeline: Standard sc-/sn-RNA-Seq Analysis in 10X Genomics data

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1 Information about this notebook

Biological heterogeneity in single-cell RNA-seq data is often confounded by technical factors including sequencing depth. The number of molecules detected in each cell can vary significantly between cells, even within the same celltype. Interpretation of scRNA-seq data requires effective pre-processing and normalization to remove this technical variability (for more information, see also scRNA-seq Dataset Integration[2]).

Hence, integration of single-cell sequencing datasets, for example across experimental batches, donors, or conditions, is often an important step in scRNA-seq workflows. Integrative analysis can help to match shared cell types and states across datasets, which can boost statistical power, and most importantly, facilitate accurate comparative analysis across datasets. Here, we include the following integration methods for integrative analysis:

(1) Seurat 'anchor-based'[3] integration workflow by Choudhary & Satija, 2022[4],
(2) Harmony[5] integration workflow by Korsunsky et al., 2019[6], and
(3) Liger[7] integration workflow by Welch et al., 2019[8].

Different integration algorithms, such as Seurat, Harmony, and Liger, often use different numbers of dimensions. The optimal number of dimensions depends on the specific algorithm and its associated parameters. While these algorithms generally consider similar variables, the mathematical approaches they use to process and transform those variables differ significantly, which means the same number of dimensions (or "resolutions") doesn't work equally well across all algorithms.

Even though they all refer to "resolution," the effect of resolution can vary from one algorithm to another, and the maximum number of dimensions they can effectively use may differ. Based on our experience, we recommend using 30 dimensions for Seurat, 30 for Harmony, and 20 for Liger. These settings tend to produce the most consistent integration results without unnecessarily increasing computational load.

2 Set up

```
attach(params)
```

```
## The following objects are masked _by_ .GlobalEnv:
##
##     integration_method, root_dir
```

```
suppressPackageStartupMessages({
  library(future)
  library(tidyverse)
  library(patchwork)
  library(Seurat)
  library(SeuratObject)
  library(harmony)
  library(rliger)
```

[2] https://www.singlecellcourse.org/scrna-seq-dataset-integration.html#seurat-v3-3-vs-5-10k-pbmc
[3] https://satijalab.org/seurat/articles/seurat5_integration
[4] https://doi.org/10.1186/s13059-021-02584-9
[5] https://portals.broadinstitute.org/harmony/
[6] https://www.nature.com/articles/s41592-019-0619-0
[7] https://welch-lab.github.io/liger/articles/liger_with_seurat.html
[8] https://doi.org/10.1016/j.cell.2019.05.006

```
  library(RcppPlanc)
  library(SeuratWrappers)
  library(scooter)
  library(reshape2)
  library(RColorBrewer)
  library(knitr)

  # Evaluate Seurat R expressions asynchronously when possible using future package
  options(future.globals.maxSize = future_globals_value)
  plan(multisession, workers = parallelly::availableCores())})
```

3 Directories and paths to file Inputs/Outputs

```
analysis_dir <- file.path(root_dir, "analyses", "integrative-analysis")
data_dir <- file.path(root_dir, "analyses", "upstream-analysis", "results", "04_Filter_obje
upstream_analysis_dir <- file.path(root_dir, "analyses", "upstream-analysis")

# Input files
data_file <- file.path(data_dir, "seurat_obj_merged_filtered.rds")

# Create results_dir
results_dir <- file.path(analysis_dir, "results")
if (!dir.exists(results_dir)) {
  dir.create(results_dir)}

# Create plots directory
report_dir <- file.path(analysis_dir, "plots")
if (!dir.exists(report_dir)) {
  dir.create(report_dir)}

plots_dir <- file.path(report_dir, glue::glue("plots_{integration_method}"))
if (!dir.exists(plots_dir)) {
  dir.create(plots_dir)}

source(paste0(root_dir, "/figures/scripts/theme_plot.R"))
source(paste0(analysis_dir, "/util/function-samples-integrate.R"))
source(paste0(analysis_dir, "/util/custom-seurat-functions.R"))
source(paste0(upstream_analysis_dir, "/util/function-process-Seurat.R"))
```

4 Read seurat object

First, we will use the `seurat_obj_merged_filtered.rds` object as generated from the pipeline in the `upstream-analysis` module. This object contains all samples of the project merged.

```
merged_obj <- readRDS(data_file)
```

5 Seurat 'anchor-based' integration workflow

Seurat integration creates a unified object that contains both original data ('RNA' assay) as well as integrated data ('integrated' assay).

5.1 Normal-sized datasets

SCT method (or sctransform) is a modeling framework for the normalization and variance stabilization of molecular count data from scRNA-seq experiments. This procedure omits the need for heuristic steps including pseudocount addition or log-transformation and improves common downstream analytical tasks such as variable gene selection, dimensional reduction, and differential expression. More specifically, the sctransform approach utilizes the Pearson residuals from negative binomial regression as input to standard dimensional reduction techniques.

5.2 Large-sized datasets

For very large datasets, the standard integration workflow can sometimes be prohibitively computationally expensive. In addition, Seurat and a lot of integration algorithms are designed to integrate data up to a maximum amount. Here, we use the `big_data` parameter as an option for integration of extremely large datasets. For example, 1.3 millions cells are too much and so this option exist for those datasets. The problem comes down to infinite matrix sizes and memory issues, no computer to data can compute this many variables so new algorithms exist to get around that through various methods. For more information, see Seurat v4 large data integration[9] and Seurat v5 large data integration[10].

A reference_list[11] can be used when conducting Seurat integration which help with speeding up integration.

Moreover, we improve efficiency and runtimes by employing (1) Reciprocal PCA (RPCA) and (2) Reference-based integration. The main efficiency improvements are gained in FindIntegrationAnchors(). First, we use reciprocal PCA (RPCA), to identify an effective space in which to find anchors. When determining anchors between any two datasets using reciprocal PCA, we project each dataset into the others PCA space and constrain the anchors by the same mutual neighborhood requirement. All downstream integration steps remain the same and we are able to 'correct' (or harmonize) the datasets.

Additionally, we use reference-based integration. In the standard workflow, we identify anchors between all pairs of datasets. While this gives datasets equal weight in downstream integration, it can also become computationally intensive. For example when integrating 10 different datasets, we perform 45 different pairwise comparisons. As an alternative, we introduce here the possibility of specifying one or more of the datasets as the 'reference' for integrated analysis, with the remainder designated as 'query' datasets. In this workflow, we do not identify anchors between pairs of query datasets, reducing the number of comparisons. For example, when integrating 10 datasets with one specified as a reference, we perform only 9 comparisons. Reference-based integration can be applied to either log-normalized or SCTransform-normalized datasets.

Note that the anchor-based RPCA integration represents a faster and more conservative (less correction) method for integration.

```
# We will use the Seurat 'anchor-based' workflow, if defined in `params`.
if (use_seurat_integration == "YES"){
```

[9] https://satijalab.org/seurat/archive/v4.3/integration_large_datasets
[10] https://satijalab.org/seurat/articles/seurat5_sketch_analysis
[11] https://satijalab.org/seurat/reference/findintegrationanchors#examples

```
  print_message <- "we use the Seurat 'anchor-based' workflow"

  # Make seurat_list
  seurat_obj_list <- list()
  seurat_obj_list <- SplitObject(merged_obj, split.by = "orig.ident")

  # Run function
  seurat_obj <- seurat_integration(seurat_obj_list = seurat_obj_list,
                                   nfeatures_value = nfeatures_value,
                                   num_dim_seurat = num_dim_seurat,
                                   num_dim_seurat_integration = num_dim_seurat_integration,
                                   big_data = big_data_value,
                                   Genome = genome_name,
                                   Regress_Cell_Cycle = Regress_Cell_Cycle_value,
                                   reference_list = reference_list_value,
                                   PCA_Feature_List = PCA_Feature_List_value)
  } else {
    print_message <- "we will skip usage of the Seurat 'anchor-based' workflow"}
```

Here, we will skip usage of the Seurat 'anchor-based' workflow.

6 Harmony integration workflow

Harmony is a general-purpose R package with an efficient algorithm for integrating multiple data sets. It is especially useful for large single-cell datasets such as single-cell RNA-seq. Harmony algorithm projects cells into a shared embedding in which cells group by cell type rather than dataset-specific conditions. Harmony simultaneously accounts for multiple experimental and biological factors. We will run Harmony within our Seurat workflow.

```
# We will use the Harmony integration workflow, if defined in `params`.
if (use_harmony_integration == "YES"){
  print_message <- "we use the Harmony integration workflow"
  seurat_obj <- harmony_integration(seurat_obj = merged_obj,
                                    variables_to_integrate = variable_value,
                                    num_dim = num_dim_harmony)
  # Plot
  name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
  p1 <- DimPlot(object = seurat_obj, reduction = glue::glue("{integration_method}"), pt.size
  p2 <- VlnPlot(object = seurat_obj, features = "harmony_1", group.by = "orig.ident", pt.siz
  print(patchwork::wrap_plots(list(p1, p2), nrow = 1))
  ggsave(file = name, width = 12, height = 5, device = "png")

  } else {
    print_message <- "we will skip usage of the Harmony integration workflow"}
```

```
## Performing Harmony Integration

## Transposing data matrix

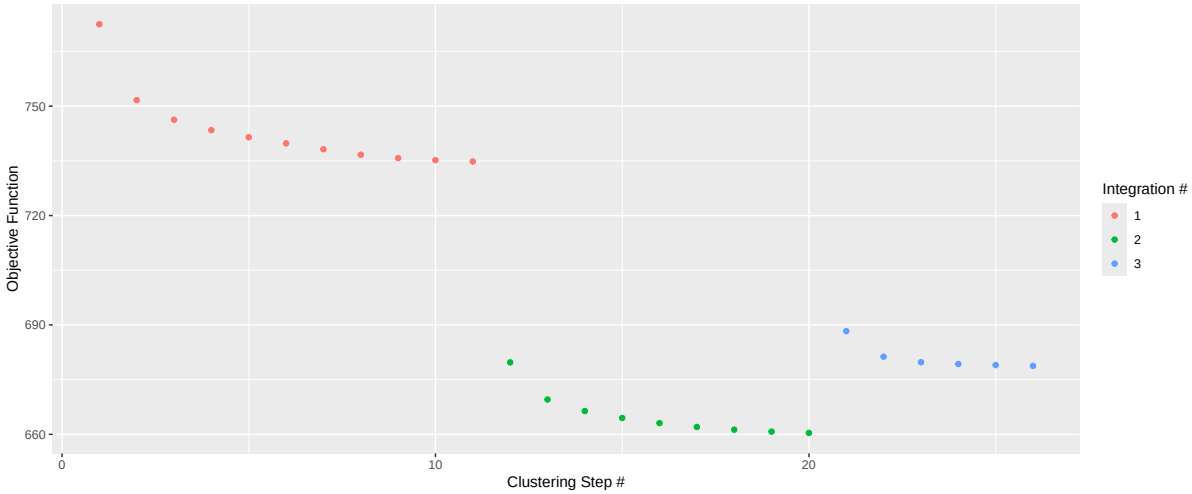
## Initializing state using k-means centroids initialization

## Harmony 1/10

## Harmony 2/10
```

```
## Harmony 3/10

## Harmony converged after 3 iterations
```




```
## Performing Uniform Manifold Approximation and Projection (UMAP) dimensional reduction tec

## 16:36:09 UMAP embedding parameters a = 0.9922 b = 1.112

## 16:36:09 Read 59358 rows and found 30 numeric columns

## 16:36:09 Using Annoy for neighbor search, n_neighbors = 30

## 16:36:09 Building Annoy index with metric = cosine, n_trees = 50

## 0%   10   20   30   40   50   60   70   80   90   100%

## [----|----|----|----|----|----|----|----|----|----|

## **************************************************|
## 16:36:16 Writing NN index file to temp file /tmp/RtmpVKrTWj/file11d9dd79f703e2
## 16:36:16 Searching Annoy index using 16 threads, search_k = 3000
## 16:36:18 Annoy recall = 100%
## 16:36:18 Commencing smooth kNN distance calibration using 16 threads with target n_neighb
## 16:36:19 Initializing from normalized Laplacian + noise (using irlba)
## 16:36:21 Commencing optimization for 200 epochs, with 2704014 positive edges
## 16:36:42 Optimization finished

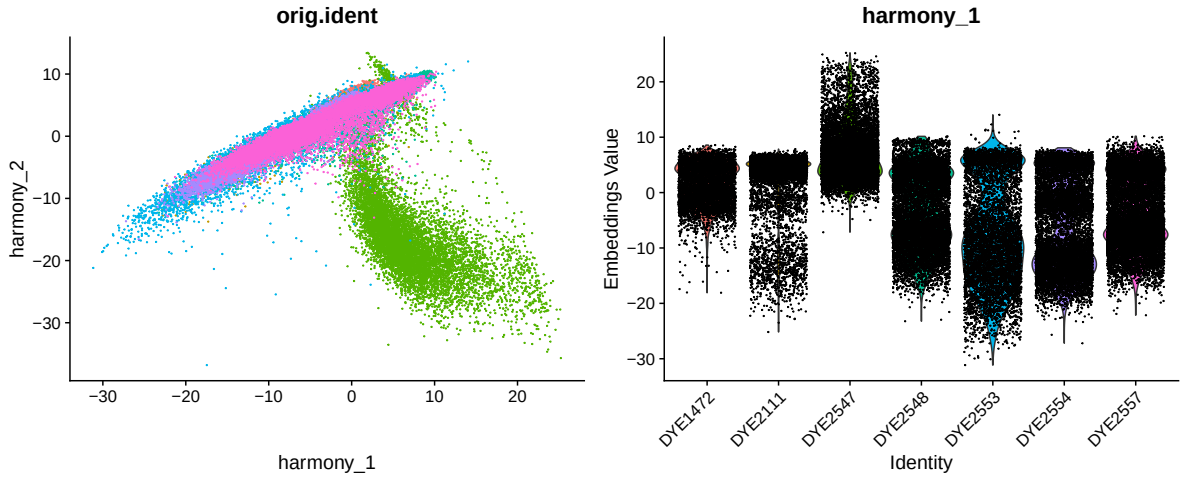
## Generate and add metadata

##
##
##  ========== saving metadata ==========
##
##
##
##
##  ========== saving metadata ==========

## Computing the k.param nearest neighbors for a given dataset and clusters

## Computing nearest neighbor graph
## Computing SNN
```

```
## Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck
##
## Number of nodes: 59358
## Number of edges: 1950539
##
## Running Louvain algorithm...
## Maximum modularity in 10 random starts: 0.8928
## Number of communities: 19
## Elapsed time: 16 seconds
## Returned Seurat object
```



Here, we use the Harmony integration workflow.

7 Liger integration workflow

LIGER is an algorithm that enables the identification of shared cell types across individuals, species, and multiple modalities (gene expression, epigenetic or spatial data), as well as dataset-specific features, offering a unified analysis of heterogeneous single-cell datasets.

More specifically, LIGER can be used to integrate datasets: (1) from multiple experiments and biological contexts (differences in technical variation, e.g., number of cells sampled, or in the depth of sequencing allocated to each cell), (2) across space (in situ and single-cell expression data to spatially locate fine subtypes of cells) and (3) various modalities (single-cell RNA-seq and DNA methylation or epigenomes). Across modalities, datasets may vary widely in dynamic range (gene expression versus chromatin accessibility), direction of relationship (RNA-seq versus DNA methylation), or in the number of genes measured (targeted quantification versus unbiased approaches).

```
# We will use the Liger workflow, if defined in `params`.
if (use_liger_integration == "YES"){
  print_message <- "we use the Liger workflow"
  seurat_obj <- liger_integration(seurat_obj = merged_obj,
                                  variables_to_integrate = variable_value,
                                  n_neighbors = n_neighbors_value)

  # Plot
  name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
  DimPlot(seurat_obj, reduction = glue::glue("{integration_method}"), group.by = c("orig.ide
```

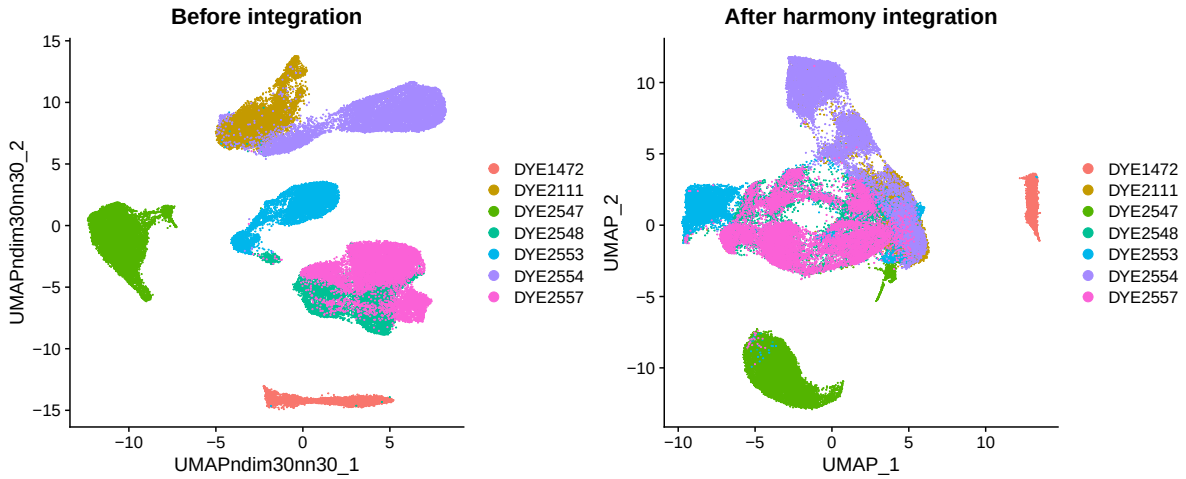
```
  ggsave(file = name, width = 6, height = 5, device = "png")

  } else {
    print_message <- "we will skip usage of the Liger workflow"}
```

Here, we will skip usage of the Liger workflow.

8 Before vs After integration plot

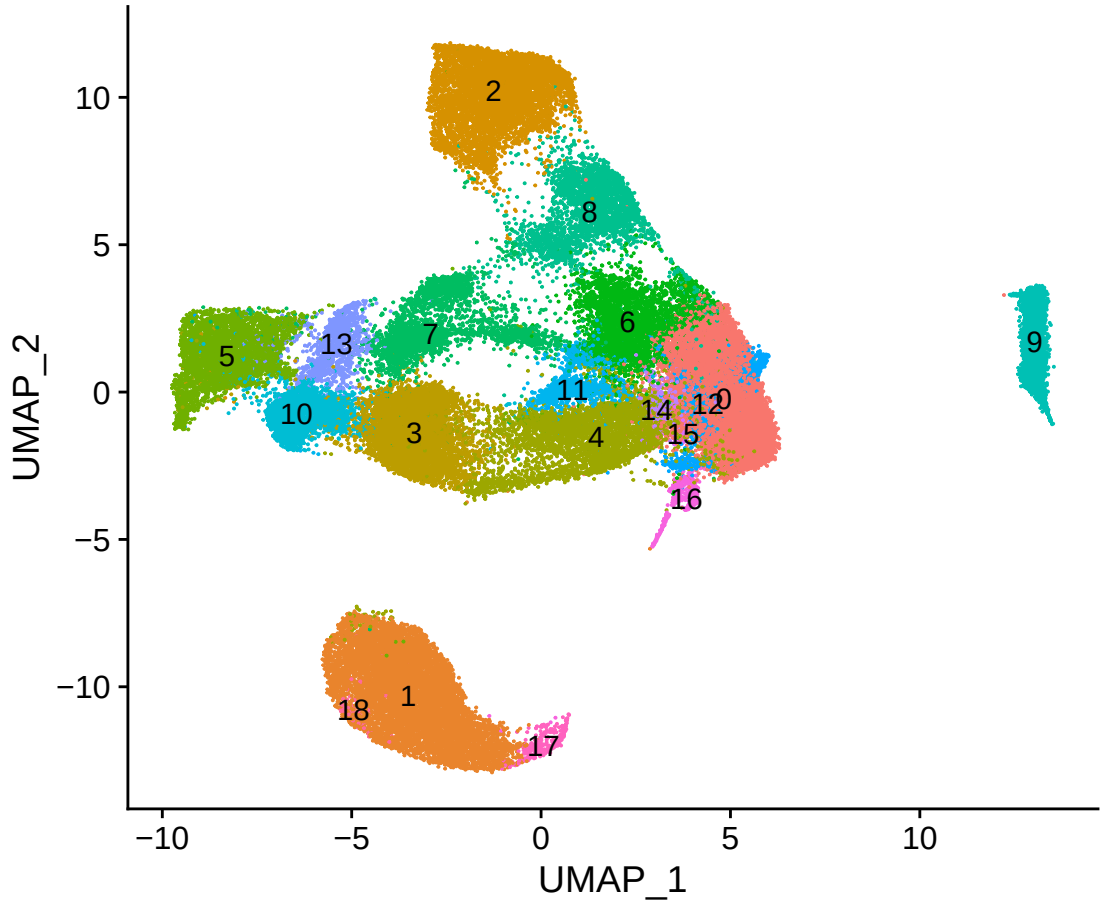
```
# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_unitegrated_vs_{integration_method}.png"))
p1 <- DimPlot(merged_obj, group.by = c("orig.ident")) + ggtitle("Before integration")
p2 <- DimPlot(seurat_obj, reduction = "umap", group.by = c("orig.ident")) + ggtitle(glue::gl
patchwork::wrap_plots(list(p1, p2), nrow = 1)
```




```
ggsave(file = name, width = 12, height = 5, device = "png")
```

```
seurat_obj <- SetIdent(seurat_obj, value = "seurat_clusters")

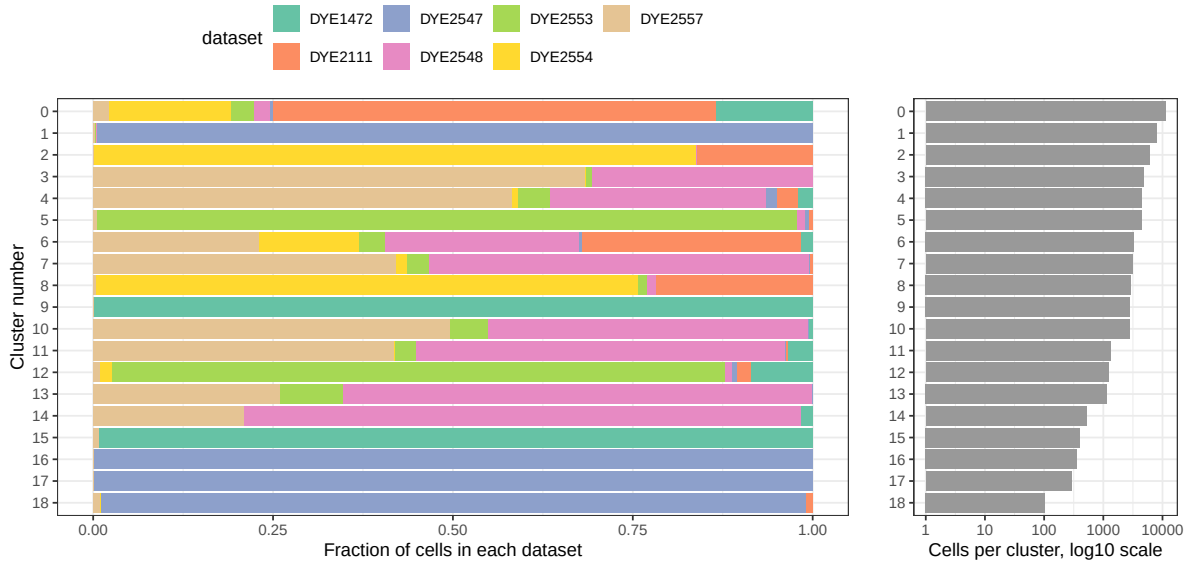
# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_clusters_{integration_method}.png"))
DimPlot(seurat_obj, reduction = "umap", label = TRUE) + NoLegend()
```




```
ggsave(file = name, width = 6, height = 5, device = "png")
```

```
# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_fraction_of_cells_{integration_method}.png")
plot_integrated_clusters(seurat_obj)
```

```
## Using cluster as id variables
```



```
ggsave(file = name, width = 10, height = 6, device = "png")
```

9 Save output files

```
# Identify columns with a '.1' suffix
cols_to_remove <- grep("\\.1$", colnames(seurat_obj@meta.data), value = TRUE)

# Exclude columns that match the specific patterns (e.g., {assay}_snn_res.0.1, {assay}_snn_
cols_to_remove <- cols_to_remove[!grepl(glue::glue("^{assay}_snn_res\\.0\\.1$"), cols_to_rem
                                 !grepl(glue::glue("^{assay}_snn_res\\.1$"), cols_to_rem
                                 !grepl(glue::glue("^{assay}_snn_res\\.10$"), cols_to_re

# Remove the columns
seurat_obj@meta.data <- seurat_obj@meta.data[, !colnames(seurat_obj@meta.data) %in% cols_to_
head(seurat_obj@meta.data)
```

```
##                          orig.ident nCount_RNA nFeature_RNA percent.mito
## DYE1472:AAACCCACAACGATTC    DYE1472       5996         3115            0
## DYE1472:AAACCCACAAGAGTAT    DYE1472       4458         2458            0
## DYE1472:AAACCCACACAGCCAC    DYE1472       2005         1330            0
## DYE1472:AAACCCACATCATCTT    DYE1472      16553         5954            0
## DYE1472:AAACCCAGTGGCCACT    DYE1472       1080          780            0
## DYE1472:AAACCCAGTTCGAAGG    DYE1472       8522         3913            0
##                          log10GenesPerUMI      ID          SAMPLE
## DYE1472:AAACCCACAACGATTC        0.9247184 DYE1472 2598223_DYE1472
## DYE1472:AAACCCACAAGAGTAT        0.9291454 DYE1472 2598223_DYE1472
## DYE1472:AAACCCACACAGCCAC        0.9460156 DYE1472 2598223_DYE1472
## DYE1472:AAACCCACATCATCTT        0.8947426 DYE1472 2598223_DYE1472
## DYE1472:AAACCCAGTGGCCACT        0.9534094 DYE1472 2598223_DYE1472
## DYE1472:AAACCCAGTTCGAAGG        0.9139987 DYE1472 2598223_DYE1472
##
## DYE1472:AAACCCACAACGATTC /research/groups/dyergrp/projects/ALSF_Pediatric_Atlas/common/EW
## DYE1472:AAACCCACAAGAGTAT /research/groups/dyergrp/projects/ALSF_Pediatric_Atlas/common/EW
## DYE1472:AAACCCACACAGCCAC /research/groups/dyergrp/projects/ALSF_Pediatric_Atlas/common/EW
## DYE1472:AAACCCACATCATCTT /research/groups/dyergrp/projects/ALSF_Pediatric_Atlas/common/EW
## DYE1472:AAACCCAGTGGCCACT /research/groups/dyergrp/projects/ALSF_Pediatric_Atlas/common/EW
## DYE1472:AAACCCAGTTCGAAGG /research/groups/dyergrp/projects/ALSF_Pediatric_Atlas/common/EW
##                                SJUID        SJID   cancer_type
## DYE1472:AAACCCACAACGATTC SJHBBBJ2DT8 SJEWS063826 Ewing_sarcoma
## DYE1472:AAACCCACAAGAGTAT SJHBBBJ2DT8 SJEWS063826 Ewing_sarcoma
## DYE1472:AAACCCACACAGCCAC SJHBBBJ2DT8 SJEWS063826 Ewing_sarcoma
## DYE1472:AAACCCACATCATCTT SJHBBBJ2DT8 SJEWS063826 Ewing_sarcoma
## DYE1472:AAACCCAGTGGCCACT SJHBBBJ2DT8 SJEWS063826 Ewing_sarcoma
## DYE1472:AAACCCAGTTCGAAGG SJHBBBJ2DT8 SJEWS063826 Ewing_sarcoma
##                          Subject_Disease_Code tissue_type
## DYE1472:AAACCCACAACGATTC                  EWS         PDX
## DYE1472:AAACCCACAAGAGTAT                  EWS         PDX
## DYE1472:AAACCCACACAGCCAC                  EWS         PDX
## DYE1472:AAACCCACATCATCTT                  EWS         PDX
## DYE1472:AAACCCAGTGGCCACT                  EWS         PDX
```

```
## DYE1472:AAACCCAGTTCGAAGG                  EWS         PDX
##                           assay_technology_updated assay_technology
## DYE1472:AAACCCACAACGATTC             10x_snRNAseq             <NA>
## DYE1472:AAACCCACAAGAGTAT             10x_snRNAseq             <NA>
## DYE1472:AAACCCACACAGCCAC             10x_snRNAseq             <NA>
## DYE1472:AAACCCACATCATCTT             10x_snRNAseq             <NA>
## DYE1472:AAACCCAGTGGCCACT             10x_snRNAseq             <NA>
## DYE1472:AAACCCAGTTCGAAGG             10x_snRNAseq             <NA>
##                           historic_data RPMI genome_reference DYExxxx
## DYE1472:AAACCCACAACGATTC           yes <NA>      dual-genome    <NA>
## DYE1472:AAACCCACAAGAGTAT           yes <NA>      dual-genome    <NA>
## DYE1472:AAACCCACACAGCCAC           yes <NA>      dual-genome    <NA>
## DYE1472:AAACCCACATCATCTT           yes <NA>      dual-genome    <NA>
## DYE1472:AAACCCAGTGGCCACT           yes <NA>      dual-genome    <NA>
## DYE1472:AAACCCAGTTCGAAGG           yes <NA>      dual-genome    <NA>
##                           DYE_Melody_unsure DYExxxx_corrected SRM SRM_Sample_ID
## DYE1472:AAACCCACAACGATTC                na           DYE1472  NA            NA
## DYE1472:AAACCCACAAGAGTAT                na           DYE1472  NA            NA
## DYE1472:AAACCCACACAGCCAC                na           DYE1472  NA            NA
## DYE1472:AAACCCACATCATCTT                na           DYE1472  NA            NA
## DYE1472:AAACCCAGTGGCCACT                na           DYE1472  NA            NA
## DYE1472:AAACCCAGTTCGAAGG                na           DYE1472  NA            NA
##                           Experimental_Sample sample_3D sample_2D Tissue_Bank_ID
## DYE1472:AAACCCACAACGATTC   EWS PDX MAST 358A        No        No    TB-17-
06841
## DYE1472:AAACCCACAAGAGTAT   EWS PDX MAST 358A        No        No    TB-17-
06841
## DYE1472:AAACCCACACAGCCAC   EWS PDX MAST 358A        No        No    TB-17-
06841
## DYE1472:AAACCCACATCATCTT   EWS PDX MAST 358A        No        No    TB-17-
06841
## DYE1472:AAACCCAGTGGCCACT   EWS PDX MAST 358A        No        No    TB-17-
06841
## DYE1472:AAACCCAGTTCGAAGG   EWS PDX MAST 358A        No        No    TB-17-
06841
##                           Xenograft_CompBio_ID CSTN_MAST_ID Date
## DYE1472:AAACCCACAACGATTC       SJEWS063826_X1     MAST358A <NA>
## DYE1472:AAACCCACAAGAGTAT       SJEWS063826_X1     MAST358A <NA>
## DYE1472:AAACCCACACAGCCAC       SJEWS063826_X1     MAST358A <NA>
## DYE1472:AAACCCACATCATCTT       SJEWS063826_X1     MAST358A <NA>
## DYE1472:AAACCCAGTGGCCACT       SJEWS063826_X1     MAST358A <NA>
## DYE1472:AAACCCAGTTCGAAGG       SJEWS063826_X1     MAST358A <NA>
##                           SRM_Submission_Information
## DYE1472:AAACCCACAACGATTC                       ALSF
## DYE1472:AAACCCACAAGAGTAT                       ALSF
## DYE1472:AAACCCACACAGCCAC                       ALSF
## DYE1472:AAACCCACATCATCTT                       ALSF
## DYE1472:AAACCCAGTGGCCACT                       ALSF
## DYE1472:AAACCCAGTTCGAAGG                       ALSF
##                           Compbioname_for_publications Comp_Bio_Sample_Name
## DYE1472:AAACCCACAACGATTC                SJEWS063826_X1                 <NA>
```

```
## DYE1472:AAACCCACAAGAGTAT               SJEWS063826_X1                 <NA>
## DYE1472:AAACCCACACAGCCAC               SJEWS063826_X1                 <NA>
## DYE1472:AAACCCACATCATCTT               SJEWS063826_X1                 <NA>
## DYE1472:AAACCCAGTGGCCACT               SJEWS063826_X1                 <NA>
## DYE1472:AAACCCAGTTCGAAGG               SJEWS063826_X1                 <NA>
##                          Comp_Bio_Subject_ID Preferred_Sample_ID
## DYE1472:AAACCCACAACGATTC            SJ063826      SJEWS063826_X1
## DYE1472:AAACCCACAAGAGTAT            SJ063826      SJEWS063826_X1
## DYE1472:AAACCCACACAGCCAC            SJ063826      SJEWS063826_X1
## DYE1472:AAACCCACATCATCTT            SJ063826      SJEWS063826_X1
## DYE1472:AAACCCAGTGGCCACT            SJ063826      SJEWS063826_X1
## DYE1472:AAACCCAGTTCGAAGG            SJ063826      SJEWS063826_X1
##                          Sample_Type_Code Prior_Comp_Bio_ID
## DYE1472:AAACCCACAACGATTC             <NA>       SJEWS063826
## DYE1472:AAACCCACAAGAGTAT             <NA>       SJEWS063826
## DYE1472:AAACCCACACAGCCAC             <NA>       SJEWS063826
## DYE1472:AAACCCACATCATCTT             <NA>       SJEWS063826
## DYE1472:AAACCCAGTGGCCACT             <NA>       SJEWS063826
## DYE1472:AAACCCAGTTCGAAGG             <NA>       SJEWS063826
##
## DYE1472:AAACCCACAACGATTC smb://jude.stjude.org/groups/dyergrp/projects/ALSF_Pediatric_Atl
## DYE1472:AAACCCACAAGAGTAT smb://jude.stjude.org/groups/dyergrp/projects/ALSF_Pediatric_Atl
## DYE1472:AAACCCACACAGCCAC smb://jude.stjude.org/groups/dyergrp/projects/ALSF_Pediatric_Atl
## DYE1472:AAACCCACATCATCTT smb://jude.stjude.org/groups/dyergrp/projects/ALSF_Pediatric_Atl
## DYE1472:AAACCCAGTGGCCACT smb://jude.stjude.org/groups/dyergrp/projects/ALSF_Pediatric_Atl
## DYE1472:AAACCCAGTTCGAAGG smb://jude.stjude.org/groups/dyergrp/projects/ALSF_Pediatric_Atl
##                          NOTES     S.Score    G2M.Score Phase    UMAP_1
## DYE1472:AAACCCACAACGATTC    NA -0.10429750  0.084216877   G2M 12.977676
## DYE1472:AAACCCACAAGAGTAT    NA -0.15771932  0.001876603   G2M 13.159720
## DYE1472:AAACCCACACAGCCAC    NA -0.01809332 -0.031313024    G1  4.873863
## DYE1472:AAACCCACATCATCTT    NA -0.01049637 -0.050616637    G1 13.096996
## DYE1472:AAACCCAGTGGCCACT    NA -0.19923618 -0.038751769    G1  4.196533
## DYE1472:AAACCCAGTTCGAAGG    NA -0.02683239 -0.139267883    G1 13.159744
##                             UMAP_2       PC_1     PC_2     PC_3       PC_4
## DYE1472:AAACCCACAACGATTC  0.5360944 -3.6866346 3.638846 6.161789 -3.5958719
## DYE1472:AAACCCACAAGAGTAT  2.9035624 -0.2570593 5.085811 5.541863 -0.3857375
## DYE1472:AAACCCACACAGCCAC -0.9161116  0.1819565 5.792733 4.749105  1.4424712
## DYE1472:AAACCCACATCATCTT  1.0134841 -4.3667239 2.121807 5.628848 -0.9009903
## DYE1472:AAACCCAGTGGCCACT -1.0832590  3.0935227 7.793377 4.581162  2.3305812
## DYE1472:AAACCCAGTTCGAAGG  1.5470963 -2.7681089 3.606610 4.988399 -1.7118416
##                                PC_5       PC_6       PC_7      PC_8        PC_9
## DYE1472:AAACCCACAACGATTC -14.486792 -10.053046 -3.0860904 -2.504382 -
1.18656511
## DYE1472:AAACCCACAAGAGTAT  -9.392747  -7.927895 -0.9519302 -0.102795  0.05390547
## DYE1472:AAACCCACACAGCCAC  -6.974901  -3.437342  0.4478024 -1.040138  0.62461973
## DYE1472:AAACCCACATCATCTT -13.835208  -9.177653 -3.5800000 -1.561639 -
0.51494453
## DYE1472:AAACCCAGTGGCCACT  -6.036866  -3.563516  1.6283616  0.608794 -
1.35881419
## DYE1472:AAACCCAGTTCGAAGG -13.297764  -8.532314 -2.6627390 -1.517763 -
1.76671348
```

```
##                                PC_10      PC_11      PC_12      PC_13
## DYE1472:AAACCCACAACGATTC  0.7418647 -0.8518244 -0.2833247  0.1374141
## DYE1472:AAACCCACAAGAGTAT -1.5596377  2.4020394 -0.7031468  1.4699164
## DYE1472:AAACCCACACAGCCAC -0.7917983  0.6490472  1.0163587 -1.4982137
## DYE1472:AAACCCACATCATCTT  1.5687359 -1.7142037 -2.2804124  1.3065883
## DYE1472:AAACCCAGTGGCCACT -1.0209973  0.5604055  0.9782135 -1.6125586
## DYE1472:AAACCCAGTTCGAAGG  0.7065814 -0.8507922 -1.3998504  0.3717175
##                                PC_14      PC_15      PC_16      PC_17
## DYE1472:AAACCCACAACGATTC  1.31524045  0.4672927  0.2559927 -0.2160838
## DYE1472:AAACCCACAAGAGTAT  1.29408247  3.2655504 -0.1898341  0.4665657
## DYE1472:AAACCCACACAGCCAC  0.03710312  2.7188721 -0.7786718  1.1990494
## DYE1472:AAACCCACATCATCTT -0.34778624 -1.5750075 -1.5461845 -0.7555571
## DYE1472:AAACCCAGTGGCCACT  0.84545928  2.4710389 -0.2083284 -1.3485224
## DYE1472:AAACCCAGTTCGAAGG  0.11985354  1.5204912 -1.2501591  0.7937947
##                                PC_18      PC_19      PC_20       PC_21
## DYE1472:AAACCCACAACGATTC -2.1401399  1.6851040  1.0047221 -0.62732035
## DYE1472:AAACCCACAAGAGTAT  0.6627130  0.8969250  0.9111651 -0.11719790
## DYE1472:AAACCCACACAGCCAC  0.5556753 -0.6086959  0.1477246 -0.24406746
## DYE1472:AAACCCACATCATCTT  0.1299417  0.8126772  0.9640505  0.76076779
## DYE1472:AAACCCAGTGGCCACT  0.7779888 -1.7807358  0.2655396 -0.63541513
## DYE1472:AAACCCAGTTCGAAGG -0.7513253 -0.7317028 -0.7933870 -0.06654824
##                                PC_22       PC_23       PC_24       PC_25
## DYE1472:AAACCCACAACGATTC -0.3562509  0.13439049 -1.52174683 -0.64818631
## DYE1472:AAACCCACAAGAGTAT -0.8566006 -0.09638485 -0.45886322 -0.36954354
## DYE1472:AAACCCACACAGCCAC  0.2815030  0.73483407 -0.38974642  0.37475651
## DYE1472:AAACCCACATCATCTT -0.1427898 -0.51723972 -0.03950362  0.39058586
## DYE1472:AAACCCAGTGGCCACT  0.3401259  0.85777977  0.31591109 -1.55415664
## DYE1472:AAACCCAGTTCGAAGG  1.1994942 -0.64185299  0.41433604  0.04772278
##                                PC_26      PC_27      PC_28       PC_29
## DYE1472:AAACCCACAACGATTC  1.2030766  0.1176553 -1.5953266 -0.31552003
## DYE1472:AAACCCACAAGAGTAT -0.9521067 -0.9763619  0.4180726 -1.44488632
## DYE1472:AAACCCACACAGCCAC  0.8685756  0.7201280  1.0361478 -0.48173373
## DYE1472:AAACCCACATCATCTT  1.1467874 -0.2982018 -0.1770549  0.44713761
## DYE1472:AAACCCAGTGGCCACT -0.3339937  0.7917786  0.1387846  0.05029187
## DYE1472:AAACCCAGTTCGAAGG -0.6083897  0.2694289 -0.6251129 -0.03914955
##                                PC_30 RNA_snn_res.0.8 seurat_clusters
## DYE1472:AAACCCACAACGATTC  1.4267086               9               9
## DYE1472:AAACCCACAAGAGTAT -0.5357271               9               9
## DYE1472:AAACCCACACAGCCAC  0.1658409              11              11
## DYE1472:AAACCCACATCATCTT  0.4328933               9               9
## DYE1472:AAACCCAGTGGCCACT  0.9323712               0               0
## DYE1472:AAACCCAGTTCGAAGG -1.0384159               9               9
```

```
saveRDS(seurat_obj, file = paste0(results_dir, "/", glue::glue("seurat_obj_integrated_{inte
```

10 Session Info

```
## R version 4.4.0 (2024-04-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK versi
##
## locale:
## [1] C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] knitr_1.49           RColorBrewer_1.1-3   reshape2_1.4.4
##  [4] scooter_0.0.0.9004   SeuratWrappers_0.2.0 RcppPlanc_1.0.0
##  [7] rliger_2.1.0.9004    harmony_1.2.3        Rcpp_1.0.14
## [10] SeuratObject_5.0.2   Seurat_4.4.0         patchwork_1.3.0
## [13] lubridate_1.9.4      forcats_1.0.0        stringr_1.5.1
## [16] dplyr_1.1.4          purrr_1.0.2          readr_2.1.5
## [19] tidyr_1.3.1          tibble_3.2.1         ggplot2_3.5.1
## [22] tidyverse_2.0.0      future_1.34.0        yaml_2.3.10
##
## loaded via a namespace (and not attached):
##   [1] jsonlite_1.8.9         magrittr_2.0.3         ggbeeswarm_0.7.2
##   [4] spatstat.utils_3.1-2   farver_2.1.2           rmarkdown_2.29
##   [7] ragg_1.3.3             vctrs_0.6.5            ROCR_1.0-11
##  [10] spatstat.explore_3.3-4 htmltools_0.5.8.1      sass_0.4.9
##  [13] sctransform_0.4.1      parallelly_1.42.0      KernSmooth_2.23-26
##  [16] bslib_0.9.0            htmlwidgets_1.6.4      ica_1.0-3
##  [19] plyr_1.8.9             plotly_4.10.4          zoo_1.8-12
##  [22] cachem_1.1.0           igraph_2.1.4           mime_0.12
##  [25] lifecycle_1.0.4        pkgconfig_2.0.3        rsvd_1.0.5
##  [28] Matrix_1.7-2           R6_2.5.1               fastmap_1.2.0
##  [31] fitdistrplus_1.2-2     shiny_1.10.0           digest_0.6.37
##  [34] colorspace_2.1-1       S4Vectors_0.44.0       tensor_1.5
##  [37] irlba_2.3.5.1          textshaping_1.0.0      labeling_0.4.3
##  [40] progressr_0.15.1       spatstat.sparse_3.1-0  timechange_0.3.0
##  [43] httr_1.4.7             polyclip_1.10-7        abind_1.4-8
##  [46] compiler_4.4.0         remotes_2.5.0          bit64_4.6.0-1
##  [49] withr_3.0.2            MASS_7.3-64            ggsci_3.2.0
##  [52] tools_4.4.0            vipor_0.4.7            lmtest_0.9-40
##  [55] beeswarm_0.4.0         httpuv_1.6.15          future.apply_1.11.3
##  [58] goftest_1.2-3          glue_1.8.0             nlme_3.1-167
##  [61] promises_1.3.2         grid_4.4.0             Rtsne_0.17
```

```
##  [64] cluster_2.1.8          generics_0.1.3         gtable_0.3.6
##  [67] spatstat.data_3.1-4    tzdb_0.4.0             data.table_1.16.4
##  [70] hms_1.1.3              sp_2.2-0               BiocGenerics_0.52.0
##  [73] spatstat.geom_3.3-5    RcppAnnoy_0.0.22       ggrepel_0.9.6
##  [76] RANN_2.6.2             pillar_1.10.1          vroom_1.6.5
##  [79] spam_2.11-1            later_1.4.1            splines_4.4.0
##  [82] lattice_0.22-6         bit_4.5.0.1            survival_3.8-3
##  [85] deldir_2.0-4           tidyselect_1.2.1       miniUI_0.1.1.1
##  [88] pbapply_1.7-2          gridExtra_2.3          scattermore_1.2
##  [91] RhpcBLASctl_0.23-42    stats4_4.4.0           xfun_0.50
##  [94] matrixStats_1.5.0      stringi_1.8.4          lazyeval_0.2.2
##  [97] evaluate_1.0.3         codetools_0.2-20       BiocManager_1.30.25
## [100] cli_3.6.3              uwot_0.2.2             systemfonts_1.2.1
## [103] xtable_1.8-4           reticulate_1.40.0      munsell_0.5.1
## [106] jquerylib_0.1.4        globals_0.16.3         spatstat.random_3.3-2
## [109] png_0.1-8              ggrastr_1.0.2          spatstat.univar_3.1-1
## [112] parallel_4.4.0         dotCall64_1.2          listenv_0.9.1
## [115] viridisLite_0.4.2      scales_1.3.0           ggridges_0.5.6
## [118] crayon_1.5.3           leiden_0.4.3.1         rlang_1.1.5
## [121] cowplot_1.1.3
```